

ENGINEERING DRAWING & TECHNICAL GRAPHICS

Comprehensive Fundamentals, Standards, Projections & Conventions

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1. Introduction to Engineering Drawing

Before manufacturing any product, object, or structure, a comprehensive design plan must be prepared. In almost all engineering workflows, the person designing the product is not the person responsible for manufacturing it. Therefore, clear, unambiguous, and standard-compliant technical information must be communicated to the production team.

Visual graphical communication acts as a universal language for technical professionals. While preliminary freehand **sketches** provide an initial qualitative overview of shape and layout, they lack the precision and quantitative data required for manufacturing. **Engineering Drawing** has evolved as a standardized, rigorous graphic medium that transmits geometric properties, dimensions, tolerances, material specifications, and surface finishes.

In Sri Lanka, standard drawing practices adhere strictly to **SLS 409 (Sri Lanka Standard Engineering Drawing Practice)**, which aligns closely with International Organization for Standardization (ISO) guidelines.

2. Engineering Drawing Categories Across Disciplines

2.1 Mechanical Engineering

Mechanical design relies on several distinct drawing formats depending on the stage of design and production:

- **Part Drawings (Detail Drawings):** Contain exact geometric details, dimensions, tolerances, and surface finishes required to manufacture an individual component.
- **Assembly Drawings:** Show how individual components fit together, including overall reference dimensions and bill-of-materials (BOM) item numbers.
- **Exploded View Drawings:** Display components separated along assembly axes in 3D perspective to assist in maintenance, repair, and clear visual ordering.
- **Development Drawings:** Layout flat pattern sheets required for folded sheet-metal components.

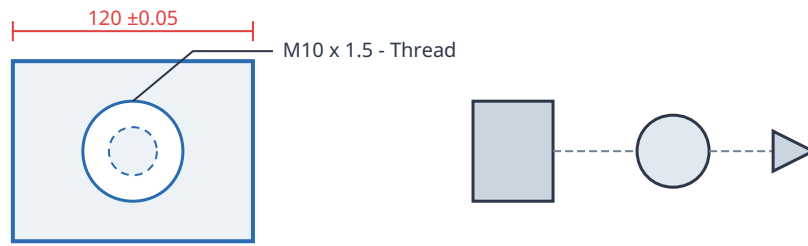


Figure 1: Mechanical Component Dimensioning & Exploded Assembly Concept

Key Mechanical Notations & Symbols:

- **M (e.g., M10):** Metric Thread designation with a nominal outer diameter of 10 mm.
- **Ø (e.g., Ø22):** Diameter indicator for cylindrical features (e.g., 22 mm diameter).
- **Tolerances (e.g., $20^{+0.02}_{-0.01}$ or ± 0.1):** Permissible limits of variation in dimension.

2.2 Civil & Construction Engineering

Civil engineering drawings specify complete structural and architectural parameters for buildings, bridges, and infrastructure. Key components include:

- **Floor Plans & Elevations:** Architectural views indicating space layouts, doors, and windows (typically drawn at 1:100 scale).
- **Cross-Sectional Views:** Cut-through views showing structural foundation depths, slab details, and roof framing (drawn at 1:20 or 1:50 scale).
- **Site & Services Plans:** Layouts for drainage systems, plumbing runs, and electrical service entries.

2.3 Electrical & Electronics Engineering

Electrical schematics utilize standardized graphic symbols to depict circuit pathways, distribution boards (DB), circuit breakers (MCBs/RCCBs), switches, and lighting layouts.

3. Drawing Equipment, Tools & Materials

3.1 Standard Drawing Paper Sizes (ISO 216 / A-Series)

Engineering papers feature high density (gsm), uniform smooth texture, and tear resistance to allow precise line work without graphite slippage.

Designation	Dimensions (mm)	Area (m²)	Typical Application
A0	841 × 1189	1.00	Large Master Site Plans, Complex Assemblies
A1	594 × 841	0.50	Civil Structural & Architectural Plans
A2	420 × 594	0.25	Building Layouts, Mechanical Assemblies
A3	297 × 420	0.125	Academic Exercises, Detail Component Drawings
A4	210 × 297	0.0625	Standard School Exercises, Data Sheets, Reports

3.2 Drawing Pencils & Grade Classification

Graphite leads are categorized from Hard (**H**) to Soft/Black (**B**):

- **Hard Grades (9H to 4H):** Extremely hard and light. Used for precise construction lines and layout grids.
- **Medium/Technical Grades (2H, H, HB, F):** Standard technical drawing pencils. **2H** is used for initial guidelines; **H** and **HB** for visible outlines and lettering.
- **Soft Grades (B to 9B):** Dark, soft graphite. Used for artistic shading and bold sketches; avoid in precision technical drawings due to smudging.

Pencil Sharpening Standard:

A standard wooden lead pencil should be sharpened to expose **8 mm** of graphite lead with a total tapered length of **24 mm to 25 mm**. Micro-lead mechanical pencils (0.35 mm, 0.5 mm, 0.7 mm) ensure constant line width without re-sharpening.

4. Standard Lines, Scales & Title Block Layout

4.1 Line Types & Conventions

Line Type	Appearance	Line Thickness	Application / Usage
Continuous Thick		0.5 - 0.7 mm	Visible object outlines and main edges
Continuous Thin		0.25 - 0.35 mm	Dimension lines, extension lines, hatching/sectioning
Dashed Thin		0.35 mm	Hidden features and concealed contours
Chain Thin		0.25 mm	Center lines, axes of symmetry, pitch circles
Chain Thick Ends		0.7 / 0.35 mm	Cutting plane locations for cross-sections
Continuous Freehand		0.25 mm	Break lines for partial views or broken sections

4.2 Standard Scales

- **Full Size (True Scale):** 1 : 1
- **Enlargement Scales:** 2 : 1, 5 : 1, 10 : 1, 50 : 1 (For micro-mechanisms, watch gears)
- **Reduction Scales:** 1 : 2, 1 : 5, 1 : 10, 1 : 20, 1 : 50, 1 : 100, 1 : 500, 1 : 1000 (For buildings, maps)

4.3 Title Block (Data Table) Specifications

The title block is placed at the bottom right corner of the drawing sheet inside the frame margin. It contains critical metadata: Title, Scale, Material, Drawing Number, Date, Designer Name, and Approver Name.

INSTITUTION / UNIVERSITY Rajarata University of RUSL	TITLE ISOMETRIC PROJECTION	DRAWING NO. DRW-2026-001
DRAWN BY V. Lakshan (Index 2798)	DATE 2026-09-17	SCALE 1 : 1
MATERIAL Mild Steel / Cast Iron	PROJECTION METHOD First Angle (ISO)	

Figure 2: Standard Engineering Title Block Layout

5. Pictorial Projection Methods

Pictorial drawings depict all three dimensions (Length, Width, Height) in a single 3D view. They are broadly divided into **Perspective Methods** and **Parallel Methods**.

5.1 Perspective Projection

Perspective projection simulates human optical vision where parallel receding lines converge at **Vanishing Points (VP)** on the Horizon Line.

- **1-Point Perspective:** Receding lines converge to a single vanishing point. Front face remains orthogonal.
- **2-Point Perspective:** Object is angled; lines converge to two separate horizon vanishing points.
- **3-Point Perspective:** Used for tall structures; lines converge to two horizon points and one vertical vanishing point.

5.2 Parallel Projection Systems

Oblique / Parallel Method:

Features horizontal and vertical principal axes. Receding depth axis is angled at 45°. Depth measurements are halved (1/2 true scale) to reduce visual distortion.

Dimetric Method:

Features two equal axis angles (7° and 42° to horizontal). Scale along the 7° and vertical axes is 1:1, while the 42° axis is foreshortened to 1/2 scale.

5.3 Isometric Projection Method

Isometric projection is the standard pictorial method in technical drawing. All three principal axes are spaced equally at 120° apart. On a horizontal baseline, the vertical axis is at 90° , while the left (X) and right (Z) axes are inclined at 30° . True dimensions are used directly along all three axes (1:1 scale ratio).

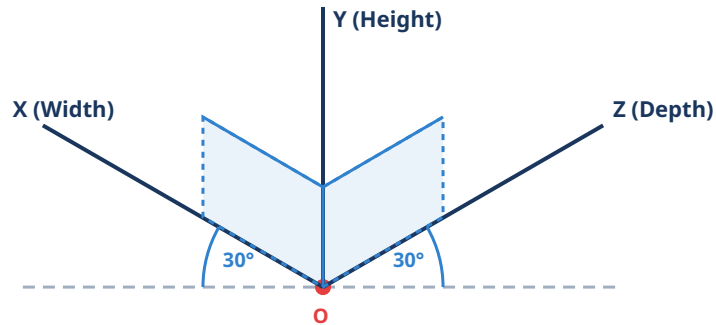


Figure 3: Isometric Axis Orientation and Geometry Axis Angles

6. Construction of Isometric Circles (Four-Center Method)

Circles projected onto isometric planes appear as ellipses. The standard geometric technique to approximate an isometric circle accurately is the **Four-Center Arc Method**:

1. Draw an isometric rhombus with side lengths equal to the diameter (D) of the circle at 30° angles.
2. Locate the midpoints of all four sides of the rhombus.
3. Draw lines from the obtuse angle vertices (120° corners) to the opposite side midpoints.
4. The intersection points of these lines with the long diagonal form two arc center points (E, G). The obtuse vertices form the remaining two center points (B, D).
5. Using these four center points, construct four tangent circular arcs to complete the smooth isometric circle.

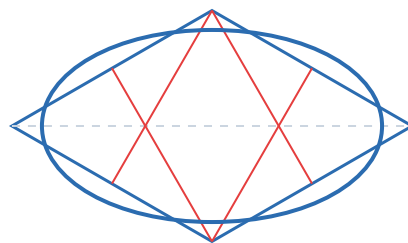


Figure 4: Four-Center Isometric Circle Construction Scheme

7. Conclusion & Practice Exercise

Mastery of engineering drawing conventions, standard lines, title block execution, and isometric projections provides the essential technical groundwork required across mechanical, civil, electrical, and technology fields.

Practice Assignment Exercise:

Using standard A4 drawing paper and a **1 : 1** scale, execute a complete engineering drawing inside a standardized frame with title block details. Draft the orthographic and isometric views of a stepped block with a cylindrical hole (Base: **100 mm**, Height: **40 mm**, Bore: **30 mm**) using proper continuous thick, dashed, and center line conventions.